

Previous Year Questions (PYQs) on Data structure

Module -1

1. Define data structure. Explain types of data structures.
2. Explain the concepts of ADT with examples.
3. Define Algorithm and list its properties.
4. Analyze the efficiency of algorithms by explaining and comparing Big O, Big Ω , and Big Θ asymptotic notations.
5. Explain time and space complexity.
6. Differentiate between
 1. Array and Linked list
 2. Stack and queue
 3. Top-down approach and bottom up approach
 4. Big-O (O), Big-Omega (Ω), Big-Theta (Θ)

Module-3

1. Define Stack and list its basic operations.
2. Define Queue. Explain different types of queue.
3. List applications of Stack.
4. Apply the concept of ADT to explain Stack operations (Push, Pop, Peek, isEmpty, isFull).
5. Describe Array representation of Stack with a neat diagram.
6. Explain Linked List representation of Stack.
7. Apply Stack operations to reverse a given string or number.
8. Convert following infix expression to prefix and postfix expression:
 - $(A-B)*(C+D)$
 - $(A+B)*C-D/E$
 - $(A+B)/(C+D)-(D*E)$
 - $A-(B/C+(D\%E*F)/G)*H$
 - $(A+(B*C))$
 - $(A-B/C)*(A/K-L)$
 - $(A+B)*C$

- $(f-g)*((a+b)*(c-d))/e$

Module -2

1. Define List Abstract Data Type (ADT).
2. List the basic operations performed on List ADT.
3. Describe Array-based implementation of List with a neat diagram.